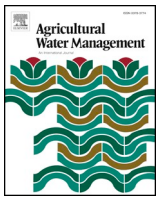
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Factors affecting crop water use efficiency: A worldwide meta-analysis



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ABSTRACT

Water is a limiting natural resource for agricultural production. While it is well accepted that crop types differ in their water use efficiency (WUE), there is no consensus on the main factors affecting WUE of main field crops, which was the motivation of the current study. The effects of rainfall pattern, soil type and climatic regime on crop WUE were evaluated using data from 514 experiments around the world published in ISI journal papers. The results confirmed that crop type had a significant effect ($p < 0.05$) on WUE with cereals producing on average 2.37 kg of dry grain per cubic meter (m^{-3}) of water followed by oilseeds (0.69 kg m^{-3}), fibre crops (0.45 kg m^{-3}) and legumes (0.42 kg m^{-3}). Amongst cereals, maize (3.78 kg m^{-3}) and sorghum (2.52 kg m^{-3}) were more water-use efficient than wheat (1.02 kg m^{-3}), barley (1.21 kg m^{-3}) and millet (0.47 kg m^{-3}). Overall, maize was the most water use efficient crop under well-watered conditions (9.90 kg m^{-3}), but sorghum was the most efficient under dry conditions (5.99 kg m^{-3}). WUE of crops increased from desert to tropical through sub-tropical climate. Moreover, WUE tended to correlate positively with soil organic carbon content ($r = 0.86$) and negatively with clay content ($r = -0.78$) and soil bulk density ($r = -0.85$). These results provide information that is important for making decisions on crop selection in a context of increased climate variability and for crop variety development with enhanced WUE. However, there is need for more research to gain more understanding of the mechanisms responsible for the observed trends and causes of the unexplained variability.

1. Introduction

Water scarcity is a major global environmental problem of the 21st century (Srinivasan et al., 2012). Globally, agriculture accounts for 80–90 % of all freshwater used by humans, and most of that water is used for crop production (Morison et al., 2008). While irrigation development has increased crop productivity in arid and semi-arid areas for decades, water scarcity and escalating costs of setting and managing the infrastructure hamper further expansion of irrigation in developing countries. Rising demand for water by other sectors such as domestic, mining, industries, the environment and, recently, severe pressures from climate uncertainties, exacerbates water shortages for further irrigation development. Therefore, there is need for new paradigms for agriculture to, at least, keep pace with rising demand for food while using the lowest possible amounts of water.

Different disciplines define water use efficiency (WUE) differently. Originally, crop physiologists defined WUE as the amount of carbon assimilated and crop yield per unit of transpiration (Viets, 1962) and later it was referred to as amount of biomass or marketable yield per unit of evapotranspiration. Irrigation scientists view WUE as a ratio of total irrigation water transpired to water diverted from the source (Israelsen, 1932), while crop scientists define it as the ratio of total biomass or grain yield to water supplied (Sharma et al., 2015). The present paper adopted WUE as the ratio of grain yield achieved to the amount of water made available to the crop (i.e. stored in the soil plus rainfall and irrigation water). It is expressed as grain yield per unit of land area (Y, kg m^{-2}) divided by the amount of water consumed by the crop per unit land area (ET, m^3 m^{-2}), usually reported as mm of water needed to produce that yield (Blum, 2005).

The need for crops to be drought tolerant and have higher WUE has been the focus of breeding efforts for many years (Sivamani et al., 2000; Tilman et al., 2002; Condon et al., 2004; Blum, 2005; Ruggiero et al., 2017) to alleviate water scarcity and food insecurity. Improved soil fertility and crop management practices have also been developed to enhance better yields with less water (Evans and Sadler, 2008; Busari and Salako, 2013; Busari et al., 2015; Wang et al., 2018). However, water scarcity remains an issue and there is still need for better understanding of the underlying controls of crop WUE.

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Table 1
Summarised database compiled using data collected from ISI journal papers showing references (authors and years) and averages of selected environmental values (mean annual precipitation and temperature, MAP and MAT respectively; number of data points (n), total amount of water onto the plot, TW; available water capacity of soil, AWC%; topsoil bulk density, BD; topsoil clay, sand and silt content, Clay, Sand and Silt, respectively) and response variables (water use efficiency, WUE; and grain yield, GY).

Crop Name	Country	n	MAP (mm y^{-1})	MAT (^oC y^{-1})	BD (g cm^{-3})	AWC	Clay	Sand	Silt	GY (t ha^{-1})	WUE (Kg m^{-3})	References
						(%)						
Cowpea	Egypt	3	30	20.70	1.50	6.32				0.39	0.62	Aboamera (2010)
Maize	Egypt	6	17	21.20	1.31	25.44				11.09	1.20	Abuarab et al (2013)
Barley, Faba bean	Saudi Arabia	16	346	15.60	1.58	7.94	3.24	81.17	15.59	9.95	0.72	Al-Neem (2008)
Sorghum	USA	8	199	21.10						1.33	1.04	Conley et al (2001)
Cotton,Maize	Turkey	20	647	17.80	1.46	20.71	11.90	44.89	26.25	5.40	1.32	Dağdelen et al (2006)
Cotton	Turkey	4	647	17.80	1.46	22.71	17.00	55.00	28.00	4.22	0.85	Dağdelen et al (2009)
Sunflower	Turkey	15	575	13.50	1.56	14.64				3.97	0.74	Erdem et al. (2001)
Maize,Wheat	China	4	580	13.50	1.35	16.00				10.72	2.36	Gao et al. (2009)
Maize	USA	12	397	19.54		12.50				10.42	1.29	
Sorghum	Egypt	12	18	21.30						43.08	5.99	Hussein and Alva (2014)
Cotton	Syria	8	198	16.90	1.16	19.00				4.09	0.66	Hussein et al. (2010)
Cotton	Uzbekistan	9	429	14.60						3.58	0.67	Ibragimov et al. (2007)
Maize	Iran	6	279.3	14.20	1.42	7.85				26.73	7.13	Kanani et al. (2016)
Wheat	China	45	542	9.00	1.20		30.90	3.50	65.60	3.57	1.09	Kang et al (2001)
Linseed,Mustard,Safflower	India	9	1472	27.00	1.58	14.50	27.23	47.17	25.60	0.66	0.19	Kar et al. (2007)
Maize	Turkey	17	679	14.00	1.40	15.00	32.80	23.60	43.60	16.79	3.45	Kuscu and Demir (2013)
Maize	Turkey	12	126.3	21.06		15.00				12.02	1.71	Kuscu et al. (2013)
Sunflower	Egypt	20	2	22.60						1.24	0.93	Mahmoud and Ahmed (2016)
Maize	Ethopia	15	831.1	20.98	1.17	17.00				5.93	10.67	Meskelu et al. (2014)
Cotton	Pakistan	2	175	25.60						2.49	0.60	Muhammad et al. (2010)
Millet	Tunisia	8	448	18.10						2.04	0.70	Nagaz et al (2009)
Cotton	Turkey	8	1109	18.10	1.43	25.33	68.93	15.15	14.93	3.55	0.74	Onder et al. (2009)
Chickpea	Syria	6	330	17.10						1.46	0.64	Oweis et al. (2004)
Maize	Italy	3	650	14.80	1.25	18.00	40.00	6.40	52.90	8.04	1.93	Paolo and Rinaldi (2008)
Maize	USA	16	508	10.60		26.00				8.98	1.45	Payero et al. (2008)
Maize	USA	16	497.86	10.91	1.58	24.87	3.24	81.17	15.59	8.98	1.50	Payero et al. (2009)
Sunflower	Pakistan	2	175	25.60						2.78	0.37	Qureshi et al. (2005)
Maize	Zimbabwe	6	850	21.50						1.72	11.81	Rusere et al. (2012)
Millet	Niger	8	546	29.00	1.65		4.00	91.00	5.00	0.89	0.24	
Cotton	USA	42	141	24.75						1.11	0.22	Snowden et al. (2013)
Maize,Sorghum,Sunflower	USA	14	419	11.10						5.68	1.18	Stone et al. (1996)
Wheat	China	4	614	13.70	1.38		15.20	2.30	82.50	4.41	1.18	Su et al. (2007),
Wheat	Turkey	22	318	11.90	1.59	14.30				5.13	1.18	Tari (2016)
Maize	Malawi	10	1142	23.80			22.50	73.80	3.90	4.80	7.60	TerAvest et al (2015)
Maize,Wheat	China	12	532	9.70	1.30					3.73	0.09	
Sorghum	USA	24	397	19.54						6.58	1.49	Tolk and Howell (2003)
Bean,Green gram	Uzbekistan	24	206	13.70		14.25				1.40	0.39	Webber et al. (2006)
Wheat	China	24	129	7.00	1.37	22.50				5.38	0.95	Xie et al. (2005)
Wheat	Syria	20	330	17.30		16.00				3.08	0.88	Zhang et al. (1998)

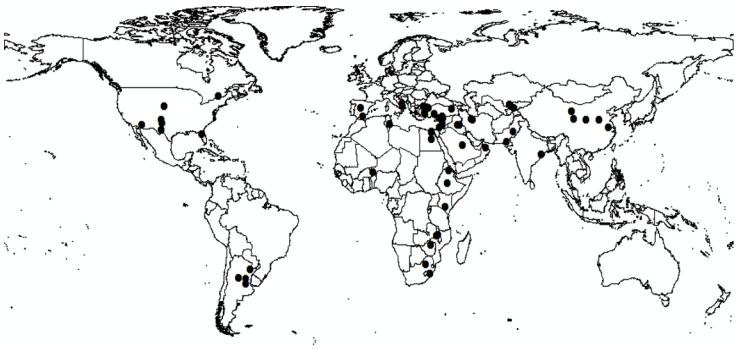


Fig. 1. Global distribution of the sites where data used in the analysis were generated.

Table 2
Definitions of the environmental factors, water use efficiency, yield and biomass as used in the analysis.

Environmental factors	Symbols	Units	Definitions
Mean annual precipitation	MAP	mm $year^{-1}$	Long-term (at least 30 year) mean precipitation per year for the study location from the papers
Mean annual air temperature	MAT	°C $year^{-1}$	Long-term (at least 30 year) mean temperature per year for the study location from the papers
Longitude	LONG	°	Longitude of the midpoint of study site as given in papers
Latitude	LAT	°	Latitude of the midpoint of study site as given in papers
Altitude	Z	m.a.s.l	Average elevation above sea level of the study site as given in the papers
Rooting depth	Zd	m	Depth within the soil profile where most of the crop roots were found
Soil bulk density	BD	g cm^{-3}	Bulk density of the top soil layer as given in papers
Total Water	TW	mm	Total amount of water received by the crop during the full crop cycle (i.e. precipitation + irrigation)
Available water capacity	AWC	%	Amount of soil water available to crops (i.e. the calculated as the difference between field capacity and permanent wilting point)
Clay content	Clay	%	Average clay content (or fine textured soil particles) of the top soils in the plot
Silt content	Silt	%	Average silt content (or medium textured soil particles) of the top soils in the plot
Sand content	Sand	%	Average sand content (or coarse textured soil particles) of the top soils in the plot
Water use efficiency	WUE	kg m^{-3}	Amount of yield per unit volume of the amount of water received in the plot
Grain yield	GY	t ha^{-1}	Grain yield yielded when the crop had matured

It is well accepted that crops differ in WUE capacities. Zwart and Bastianssen (2004), in a global meta-analysis, found WUE to increase from 0.6–1.7 kg m^{-3} for wheat, 0.6–1.6 kg m^{-3} for rice to 1.1–2.7 kg m^{-3} for maize. Several other studies pointed that cereals tend to have higher WUE than oilseed crops (Norton and Waschsmann, 2006; Sadras and Mcdonald, 2012). Water supply also showed significant effect on crop WUE, with several studies reporting a general increase of WUE with decreasing water supplied (e.g. Erdem et al., 2001; Rusere et al., 2012; Mabhaudhi et al., 2013; Chibarabada et al., 2015). For example, Erdem et al. (2001) reported higher sunflower WUE under water stressed than under well-watered conditions in their study at Tekirdag, Turkey. In South Africa, Chibarabada et al. (2015) reported the highest Bambara groundnut WUE under severely water stressed conditions (WUE of 0.006 g mm^{-1} at 25 % Field Capacity, FC) and lowest WUE under non-stressed conditions (0.003 g mm^{-1} at 75 % FC). Rusere et al. (2012) also reported increasing WUE with decreasing amount of water applied in irrigation based winter silage maize trials in Zimbabwe. Type of irrigation system used was also shown to have significant effect on WUE with drip systems yielding higher cotton WUE (0.77 kg m^{-3}) than the furrow systems (0.49 kg m^{-3}) (Ibragimov et al., 2007).

In addition, other factors such as soil type, climate, crop and land management practices also exhibit significant effects on crop WUE. Ismail and Ozawa (2007) reported from 45 to 64 % higher WUE in crops grown on clayey than sandy soils. In a different study, Dou et al. (2016) also reported (25 %) higher WUE for rice grown on clayey than sandy loam soils. Reported higher WUE for wheat, oats, potato and maize under warm-dry than warm-wet climate in northern China. Fan et al. (2005) reported higher WUE in fertilized (0.95 kg ha^{-1} mm^{-1} and 0.96 kg ha^{-1} mm^{-1} for wheat and maize respectively) than un-fertilized crops (0.32 kg ha^{-1} mm^{-1} and 0.47 kg ha^{-1} mm^{-1} for wheat and maize respectively) in Gansu, China. Moreover, Wang et al. (2018) reported an increase of water use efficiency of winter wheat by 5.1 %, 13.8 % and 29.3 % respectively for the application rate of 60, 120 and 180 kg N ha^{-1} respectively. Zhang et al. (2017) pointed to a combined effect of water and nitrogen applications and showed that a decreasing water supply with nitrogen fertilizer application before cotton flowering and increasing the amounts at later growth stages increase WUE. Sinclair (2018) gives some explanation on how water shortage may improve WUE. This author showed that drought occurring early in the crop growth cycle partially closes stomata which results in the conservation of soil water and in a subsequent improved crop yield per unit of water.

Conservation tillage has also the ability to enhance WUE in comparison to conventional tillage in Henan Province, China. In addition, mulching of crop residues was reported to reduce evapotranspiration (Xie et al., 2005; Zheng et al., 2018) with, for instance, Xie et al. (2005) reporting 16 % higher WUE in mulched than un-mulched wheat (0.875 and 1.02 kg m^{-3}, respectively).

Several studies have reported disparities in crop WUE under different soil types, climates and water regimes, and there is no consensus yet on the impact of these factors. However, the existence of studies across the world provides an opportunity for a comprehensive analysis to draw a general understanding of the main controls of crop WUE. Therefore, the objectives of the current study were to compile and analyze available studies on the WUE of main crops in order to assess and quantify variations due to soil and environmental conditions. In pursuit of these objectives, the study compiled data from 514 experiments around the world.

2. Methods and materials

2.1. Database preparation

Online sources of published ISI journal papers sources such as ISI

Table 3
List of classes describing the environmental conditions and crop types as used in the analysis.

Environmental factors	Remarks	Class range	Name
Climate (MAP, mm year^{-1}; MAT, °C year^{-1})	Hot and wet	MAT > 20 MAP > 1000	Tropical Desert
	Hot and dry	MAT > 15 MAP:0–100	
	Warm and arid-humid	MAT: 10–20 MAP: < 100–1110	Subtropical
	Cool and arid-moist	MAT < 10 MAP: 120–1000	Temperate
Clay (%)	Average clay content of the top soil horizon	0–20 20–35 > 35	Low Medium High
Sand (%)	Average sand content of the top soil horizon	0–25 25–50 > 50	Low Medium High
Silt (%)	Average silt content of the top soil horizon	0–20 20–40 > 40	Low Medium High
Total water (TW, mm)	Amount of water received by a plot (sums of precipitation and irrigation water applied)	0–500 500–1000 > 1000	Low Medium High
Crop type	Field grain crops	wheat, maize, sorghum, millet, barley	Cereal
	Field legume crops	Beans, peas	Legume
	Field oil seed crops	linseeds, mustard, safflower, sunflower	Oilseed
	Field fibre crops	cotton	Fibre
Growing season	Field crops grown under the different seasons	wheat, barley Maize, sorghum, pearl millet	Winter Summer

Some of the factor classes were adapted from.

Web of Science, Science Direct, Springerlink, Scopus, Google Scholar and ResearchGate were searched for literature on crop WUE, from March to July 2017. Key terms used to facilitate the search include "water", "use efficiency", "water use efficiency". Only papers reporting on WUE for field trials and indicating the trial treatments and environmental conditions (e.g. climate, soil type, amount of rainfall or water applied) were accepted. In addition to grain yield, aboveground biomass data were captured when available. The final database, summarised in Table 1, consisted of 512 WUE data points from 60 peer-reviewed ISI journal papers. Fig. 1 shows the locations of the data points. The main data contributors were USA (n = 132), Turkey (n = 98), China (n = 89), Egypt (n = 41) and Syria (n = 34). Southern Africa had Malawi (n = 10) and Zimbabwe (n = 6).

2.2. Definitions of WUE and environmental factors

In the current study, WUE is defined as the amount of grain yield produced per unit volume of water used (kg m^{-3}) regardless of growing season and crop type or cultivar. Grain yield is the amount of grains produced per crop. Table 2 presents the definitions of selected soil and environmental conditions. The soil parameters are topsoil clay, sand and silt content, bulk density, and amount of water at field capacity, permanent wilting point and available water capacity. The other environmental factors related to site location (LONG: longitude, LAT: latitude), altitude (Z), mean annual precipitation (MAP), mean annual air temperature (MAT).

When climatic characteristics were not available in the papers, surrogate data from nearby locations (e.g. towns) were used. All these variables were categorized; information on the classes is given in Table 3. Several types of climates were derived from MAP and MAT data following Mathew et al. (2017). Tropical represents hot (MAT > 20 °C year^{-1}) and wet (MAP > 1000 mm year^{-1}) climate; subtropical depicts warm (MAT: 10–30 °C year^{-1}) and arid to humid (MAP: 100–1110 mm year^{-1}); temperate represents cool (MAT < 10 °C year^{-1}) and arid to moist (MAP: 120–1000 mm year^{-1}); while desert corresponds to warm (MAT > 15 °C year^{-1}) and dry (MAP: 0–100 mm year^{-1}) zone. Soil texture (clay, sand and silt content) and amount of water supplied (precipitation plus irrigation) were categorised into low, medium and high classes. Crops were categorised by type and growing season.

2.3. Data analyses

Descriptive statistics (minimum, maximum, median, mean, SEM: standard error of mean, quartile 1 and quartile 3 representing 25th and 75th percentiles, respectively, skewness (Skew), kurtosis (Kurt) and coefficient of variation (CV%) were calculated for all study variables (Table 4). Mean WUE values were computed for different crop types, crop growing seasons and environmental factor classes. Significance differences between factor class values were tested at $p < 0.05$ using t-tests.

3. Results

3.1. General statistics of soil and environmental variables

The type of environmental and climatic conditions varied widely, with MAP (mean 420 ± 13 mm y^{-1}) ranging from 2 mm y^{-1} in Egypt where sorghum production was under sole irrigation (Hussein and Alva, 2014) to 1472 mm y^{-1} in India (Kar et al., 2007)(Table 4). The lowest irrigation water was 12 mm applied to wheat in Turkey (Tari, 2016), while the highest amount was 7960 mm applied to sorghum in Egypt (Hussein and Alva, 2014). The average supplementary irrigation was 510 ± 51 mm. MAT (mean = 16.5 ± 0.3 °C y^{-1}) showed less variation than MAP (CV of 34 and 70 % for MAT and MAP, respectively) and ranged from 7.0 °C y^{-1} in China (Xie et al., 2005) to 29.0 °C y^{-1} in

Table 4
Descriptive statistics (Min: minimum, Max: maximum, Qt1 and Qt3: quartile 1and quartile 3, respectively, SEM: standard error of mean, Skew: skewness, Kurt: kurtosis, CV%: coefficient of variation) of environmental factors and field crop variables (water use efficiency and grain yield) included in the global database.

Variables	n	Mean	Median	Min	Max	Qt1	Qt3	SEM	CV%	Skew	Kurt
AWC	282	17.29	16.00	6.32	26.00	14.30	22.50	0.32	31	016	−0,75
Altitude	514	721.10	861.00	4.00	1602.00	93.00	1170.00	23.21	72	−0,04	−1,35
BD	252	1.39	1.40	1.16	1.65	1.20	1.58	0.01	11	−0,02	−1,31
Biomass	243	1182.00	909.30	27.13	5950.00	421.00	1645.00	66.48	88	1,60	3,34
Clay	145	24.44	30.90	3.24	68.93	15.20	30.90	1.30	64	0,81	1,42
FC	252	30.53	32.00	12.70	51.25	24.80	35.20	0.51	26	0,18	0,34
GY	514	6.20	4.05	0.21	53.00	1.82	7.11	0.33	122	3,37	14,08
Irrigaion treat	490	510.40	278.00	12.00	7960.00	121.00	488.00	50.82	220	5,48	30,61
K-rate	99	68.59	11.25	0.00	840.00	0.00	60.00	16.62	241	4,14	16,54
LAI	53	13.66	3.70	0.80	103.30	2.65	4.79	3.55	189	2,30	3,89
Latitude	514	35.33	35.23	−17.64	345.00	33.07	38.93	1.31	84	8,72	90,33
Longitude	514	18.74	31.35	−112.00	113.90	2.28	71.75	3.22	389	−0,66	−0,80
MAP	514	420.50	397.00	2.00	1472.00	198.00	542.00	12.91	70	1,25	2,63
MAT	514	16.51	17.00	7.00	29.00	11.90	21.06	0.25	34	0,15	−0,95
N-Rate	246	149.30	126.20	36.00	600.00	50.00	169.00	8.93	94	2,20	4,68
OM	157	1.51	1.55	0.00	4.78	1.15	1.64	0.08	64	1,56	4,43
P-rate	206	91.49	46.00	0.00	430.00	21.00	80.00	8.18	128	2,00	2,98
PlantPop	362	188.60	16.00	3.64	3381.00	7.14	200.00	31.94	322	4,82	22,41
Plot size	422	123.10	33.60	1.00	1000.00	15.00	225.00	8.98	150	2,50	7,59
Precipitation	380	243.00	239.60	5.00	1169.00	73.00	330.00	10.68	86	1,67	4,60
RD	165	1.33	1.20	0.50	2.00	0.90	1.83	0.04	39	−0,03	−1,38
RH	143	59.96	59.50	48.44	70.00	56.65	63.90	0.47	9	0,19	−0,81
SK	93	422.20	176.00	75.00	1086.00	165.00	544.00	37.10	85	0,92	−0,56
SOC	33	0.66	0.34	0.20	1.40	0.31	1.03	0.08	67	0,44	−1,29
SON	111	357.10	0.11	0.00	1100.00	0.11	968.80	44.67	132	0,59	−1,64
SP	142	14.10	13.44	0.10	59.00	0.10	15.00	1.52	128	1,70	1,76
Sand	145	35.15	23.60	2.30	91.00	3.50	55.00	2.59	89	0,47	−1,28
Silt	145	38.02	28.00	3.40	82.50	15.59	65.60	1.97	62	0,14	−1,41
Total Water	514	669.90	447.80	0.00	7960.00	320.00	617.60	47.84	162	5,43	30,78
WP	234	13.46	11.20	6.38	26.25	9.00	17.50	0.35	40	0,72	−0,67
WUE	514	1.74	1.04	0.03	14.80	0.60	1.50	0.11	143	3,02	9,19
Water-used	84	544.80	459.50	169.00	1304.00	360.00	665.50	29.89	50	1,23	1,11

MAP = mean annual precipitation (mm yr^{-1}); MAT = mean annual air (°C); Z = altitude (masl); RH% = relative humidity (%); BD = top soil bulk density (g cm^{-3}); Clay, Sand and Silt for top soil clay, sand and silt content (%); Population = plant population (plants m^{-2}); Irrigation = amount of irrigation water applied (mm); Total water = Sum of precipitation and irrigation (mm),Water used = amount of water used by crop (mm); ETc = Crop evapotranspiration for the entire crop cycle (mm); AWC% = available water capacity of soils (%); FC = field capacity (%),WP = permanent wilting point (%); Zd = rooting depth of crops (m); LAI = leaf area index; N, P and K rate = application rate of N, P and K (kg ha^{-1}); Biomass = above ground biomass (g m^{-2});GY = grain yield (t ha^{-1}) WUE = water use efficiency (kg m^{-3}).

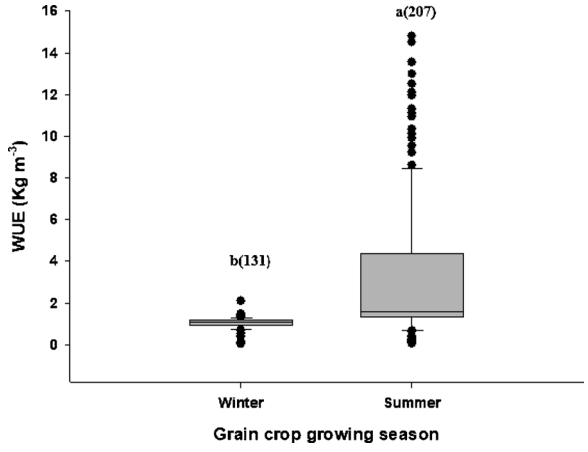


Fig. 2. Box plots comparing WUE (kg m^{-3}) variations for field grain crops grown in winter and summer used in the analysis. Box plots accompanied by similar letters were not significantly different at $p < 0.05$. Numbers between brackets are the sample sizes. See Table 3 for the crops in each season class.

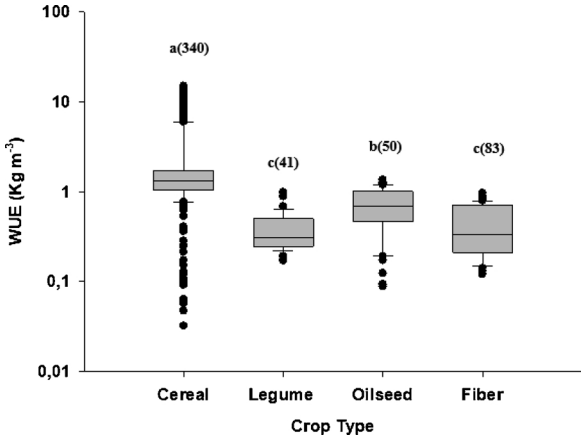


Fig. 3. Box plots comparing WUE (kg m^{-3}) variations among field crop types/ categories used in the analysis. Box plots accompanied by similar letters were not significantly different at < 0.05. Numbers between brackets are the sample sizes. Y-axis is in logarithmic scale. See Table 3 for the crop type classes.

Niger (Sivakumar and Salaam, 1999). Relative humidity (59.96 % ± 0.5 %) showed even lower variation (CV = 9 %) with values from 48.4 % in Syria (Hussein et al., 2010) to 70 % in Turkey (Onder et al., 2009). Bulk density of top soil (1.39 ± 0.01 g cm^{-3}) ranged from 1.16 g cm^{-3} in Damascus, Syria (Hussein et al., 2010) to 1.65 g cm^{-3} in Niger (Sivakumar and Salaam, 1999). Top soil sand content (35.2 ± 2.5 %) exhibited the highest variation amongst textural properties (CV = 89 %) with minimum value of 2.30 % found in China (Su et al., 2007) and a maximum of 91.0 % in Niger (Sivakumar and Salaam, 1999). Top soil clay content (mean = 24.4 ± 1.3 %; CV = 64 %) varied from 3 % in Saudi Arabia (Al-Neem, 2008) to 69 % in Turkey (Onder et al., 2009).

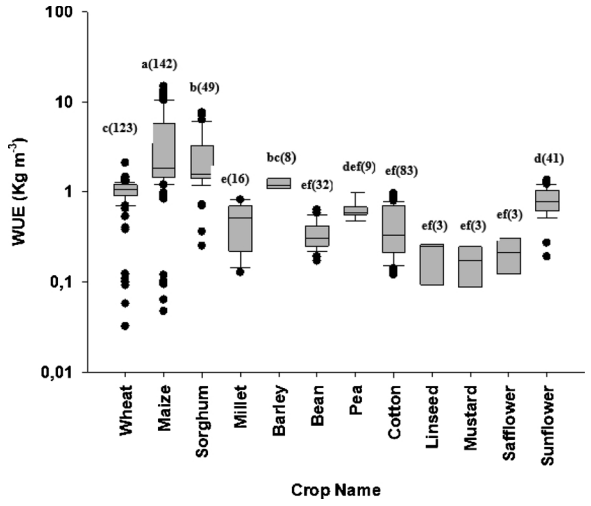


Fig. 4. Box plots comparing WUE (kg m^{-3}) variations among the different field crops used in the analysis. Box plots accompanied by similar letters were not significantly different at < 0.05. Numbers between brackets are the sample sizes. Y-axis is in logarithmic scale.

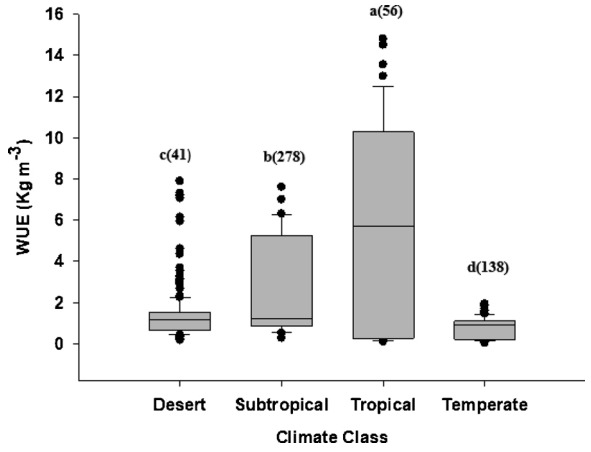


Fig. 5. Box plots comparing WUE (kg m^{-3}) variations for field crops under different climatic regions. Box plots accompanied by similar letters were not significantly different at $p < 0.05$. Numbers in brackets are the sample sizes. See Table 3 for climate class.

3.2. Impact of crop type on WUE

The mean WUE was 1.74 ± 0.11 kg m^{-3} with values ranging from 0.03 kg m^{-3} for wheat in China to 14.8 kg m^{-3} for maize in Ethiopia (Meskelu et al., 2014) (Table 4). Summer crops had higher WUE (average of 3.23 kg m^{-3}) than winter crops (1.03 kg m^{-3}), a 3.13 times difference, significant at $P < 0.05$ (Fig. 2). Cereals had the highest WUE (2.37 kg m^{-3}) amongst the cultivated crop types studied (Fig. 3). WUE decreased from cereals to oilseed crops (0.69 kg m^{-3}), fibre crops (0.45 kg m^{-3}) and to legumes (0.45 kg m^{-3}), all the differences were significant at $P < 0.05$.

Amongst cereals, maize (mean of 3.78 kg m^{-3}) had 1.5 fold greater WUE than sorghum (2.52 kg m^{-3}) (Fig. 4). The WUE decreased to barley (1.21 kg m^{-3}), wheat (1.02 kg m^{-3}) and to pearl millet (0.47 kg m^{-3}). Linseed (0.21 kg m^{-3}) and mustard (0.17 kg m^{-3}) had the lowest WUE in the data set (Fig. 4).

3.3. Impact of environmental and soil conditions on WUE

3.3.1. Climate

Fig. 5 shows that WUE increased with increasing precipitation from desert (mean WUE of 1.34 kg m^{-3}) to subtropical (2.43 kg m^{-3}) and to tropical (5.69 kg m^{-3}). The increase in WUE was of 81 %, from desert to subtropical and by 134 % from subtropical to tropical. In deserts (Fig. 6A), sorghum had the highest WUE (5.99 kg m^{-3}) and was followed by maize (1.20 kg m^{-3}), sunflower (0.93 kg m^{-3}) and pea (0.62 kg m^{-3}). Under subtropical conditions (Fig. 6B), maize had the highest WUE (2.32 kg m^{-3}) and was followed by sorghum (1.49 kg m^{-3}). Surprisingly, WUE of barley and wheat did not significantly differ, as this of sunflower, millet, cotton and pea (Fig. 6B). Under the tropical climate, maize had significantly higher WUE (9.90 kg m^{-3}) which was 9.5 times higher than for sorghum (1.04 kg m^{-3}) whilst WUE of millet, safflower, linseed and mustard were all below 0.3 kg m^{-3} (Fig. 6C). Under temperate climate (Fig. 6 D), maize had the highest WUE (1.17 kg m^{-3}), but was not significantly different from wheat (0.97 kg m^{-3}), sorghum (0.4 kg m^{-3}) and cotton (0.24 kg m^{-3}), probably owing to low number of data points. The increase of WUE with precipitation was confirmed by correlation coefficient (r) of 0.52 displayed in Table 5.

3.3.2. Soil texture

Clay, sand and silt correlated negatively with crop WUE. The correlations coefficients were −0.89, −0.40 and −0.40 for clay, sand and silt respectively (Table 5). Box-plots also showed significant soil texture effect on WUE (Fig. 7). Field crop WUE was generally higher under loam textured soils (3.13 kg m^{-3}), despite a greater variability, than at the other textures. However, WUE between loam and clay soils were not significantly different at about 2 kg m^{-3}but WUE was significantly lower under clay loam (1.27 kg m^{-3}) and silt loam (1.19 kg m^{-3}) conditions (Fig. 7).

Fig. 8 revealed that maize was the most water use efficient crop under clay (WUE of 10.67 kg m^{-3}, Fig. 8A), loam (9.45 kg m^{-3}, Fig. 8B) and clay-loam (2.31 kg m^{-3}, Fig. 8D) but was the second highest under silt-loam. Wheat had the second highest WUE on clay (1.18 kg m^{-3}) and loam soil (1.10 kg m^{-3}) (Fig. 8A and B). Sorghum (1.48 kg m^{-3}) and maize WUE (1.47 kg m^{-3}) were not significantly different on silt loam soil; while cotton had the lowest WUE (0.22 kg m^{-3}) on clay loam soil (Fig. 8D).

4. Discussion

4.1. Impact of crop type on WUE

Water use efficiency (WUE) varied with crop type, with our results indicating that maize and sorghum tended to have higher WUE than other members of the grass family such as wheat and barley (Fig. 3). The significant differences in WUE between maize and sorghum on one hand and wheat on the other hand may be explained by differences in their photosynthetic pathways. These two groups have different photosynthetic pathways linked to the way they evolved genetically. The photosynthetic pathways have been used to classify plants into C4 and C3 types. The C4 plants (which include sugarcane, maize, sorghum and millet) originated in subtropical areas, while C3 plants (which include wheat, barley and rice) populated a far broader range of climates (Yamori et al., 2014). Studies have generally shown that C4 plants tend to be more water use efficient than C3 species under both natural and managed ecosystems (Blankenagel et al., 2018; Way et al., 2014; Zwart and Bastiaanssen, 2004). The higher WUE in C4 than C3 crop types is generally attributed to increased CO_2 concentration mechanisms in the special bundle sheath cells of C4 crop types, which allow higher photosynthetic rates while reducing stomatal conductance. The reduced stomatal conductance reduces transpiration, resulting in water savings (Sage and Monson, 1999). However, whether C4 are more water use efficient than C3 plants or not appears to be controlled by the growing environment; for example, C4 plants tend to outperform C3 plants in photosynthetic prowess under hot and dry conditions, while C3 plants perform better under sufficient water and sunlight conditions.

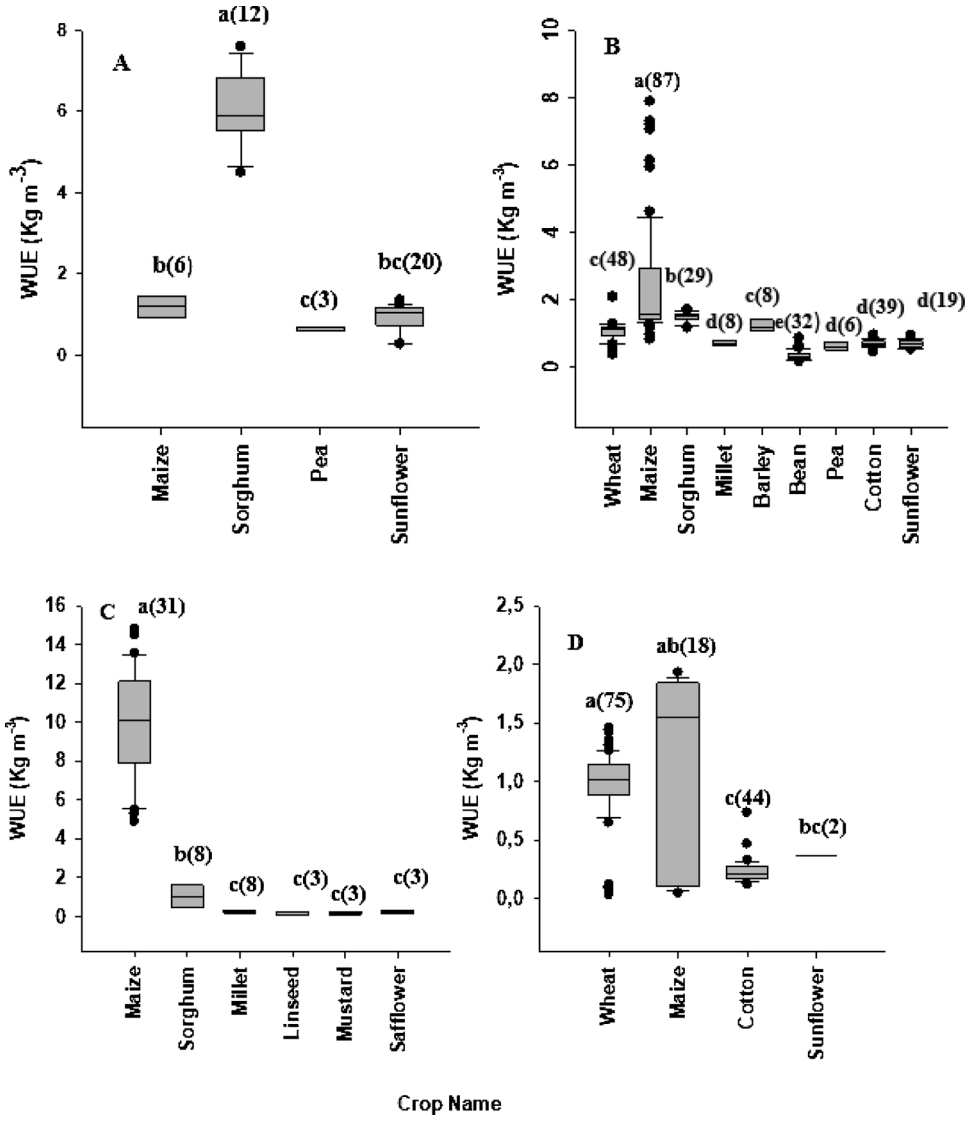


Fig. 6. Box plots comparing WUE (kg m^{-3}) variations amongst field crops grown in different environments (A) Desert, (B) Subtropical (C) Tropical and (D) Temperate. Box plots accompanied by similar letters were not significantly different at $p < 0.05$. Numbers in brackets are the sample sizes.

The growing conditions used in the current analysis appear to have favoured C4 plants more than C3 plants. These findings agree with results from other studies. For example, Katerji and Mastorilli (2014) who found higher WUE in sorghum than other major grain crops including wheat and barley in a Mediterranean environment. The impact of growing environment may also explain the low WUE for pearl millet, a C4 plant type, which was even lower than wheat WUE. Pearl millet is one of the least popular cereal crops grown worldwide but is highly adapted to hot and dry conditions, where research efforts are generally limited. Hence, research related activities around this crop tend to occur in environments less favourable for its optimum performance.

The current results also showed cereal crops to have the highest WUE (Fig. 2), which was largely anticipated because the group was dominated by C4 crops. The WUE tended to decrease from cereal crops, to oilseed crops, fibre crops and to legumes, which was also expected because other studies elsewhere showed a similar trend. Angadi et al. (2008), who evaluated WUE of different crop types, found that wheat (cereal crop) had the highest WUE followed by pulses (legumes) and oilseeds having the lowest WUE. Legumes are C3 plants and would be expected to have lower WUE than cereal crops dominated by C4 plants. It is also generally known that yield per unit area basis tends to decrease as crops are improved for such properties as high proteins; oil and fibre production and this might explain the general decrease because WUE was on the basis of yield. The lack of significant difference between oilseeds and fibre crops may be attributed to higher variability of WUE under the oilseeds.

The result that summer grain crops had significantly higher WUE than winter grain crops (Fig. 4) was reflective of the performance of C4 vs. C3 plants. The summer crop group consisted of maize, sorghum and millet which are C4 type, while the winter group was composed of wheat and barley which are C3. Summer conditions might also be more conducive for crop productivity especially in the southern hemisphere where summers are characterised by relatively higher temperatures, soil moisture and more sunshine hours. Southern hemisphere winters are generally cooler and drier with less sunshine hours. However, high variability of WUE for both groups might be indicative of the existence of other critical controls of WUE. Thus, the impacts of selected environmental controls were also elucidated.

4.2. Impact of environmental conditions on WUE

4.2.1. Climate

Climatic factors (MAP and MAT) showed significant effects on WUE with a tendency for WUE to increase from desert-like environments to

Table 5
Spearman rank correlation coefficients between selected controlling factors and WUE.

Variables	WUE
AWC%	0.333*
Altitude	−0.558**
BD	−0.736**
Biomass	0.577**
Clay	−0.894**
FC%	−0.894**
GY	0.834**
Irrigation	0.150
K rate	−0.319*
LAI	0.097
Latitude	0.816**
Longitude	−0.816**
MAP	−0.558**
MAT	−0.816**
N rate	0.894**
OM	0.025
P rate	−0.457**
PlantPop	−0.515**
Plot size	0.494**
Precipitation	0.523**
RD	0.070
RH%	−0.783**
SK	0.745**
SOC	0.781*
SON	0.534**
SP	0.516**
Sand	−0.402*
Silt	−0.402*
Tot water	0.177*
WP%	−0.327
Water used	−0.027

MAP = mean annual precipitation; MAT = mean annual air temperature; RH% = relative humidity of the atmosphere; OM = soil organic matter content; SOCc = soil organic carbon content; SONc = soil organic nitrogen content; Soil P = soil phosphorus content; Soil K = soil potassium content; Total water = some of rainfall and irrigation water applied to a crop; AWC% = available water capacity of the soil; FC% = field capacity; WP% = permanent wilting point; LAI = leaf area index; ETc = evapotranspiration of the crop, Precipitation = amount of rainfall and snow received during the growing season.

* significant at $p < 0.05$,** significant at $p < 0.01$.

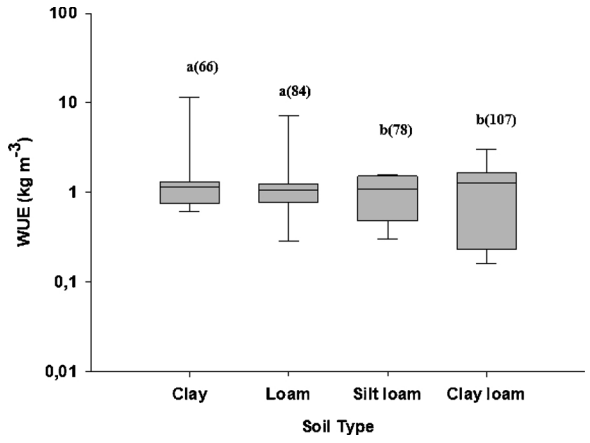


Fig. 7. Box plots comparing field crops WUE (kg m^{-3}) variations under different soil types. Box plots accompanied by similar letters were not significantly different at $p < 0.05$. Numbers in brackets are the sample sizes. Y-axis in logarithmic scale.

subtropical and to tropical zones. Maize and sorghum exhibited higher WUE in desert, subtropical and tropical environments (Fig. 6A–C). This was most likely to be due to the fact that both maize and sorghum are classified as C4 crops and have been shown to exhibit higher drought tolerance than other crops (Chipanshi et al., 2003). The increase of WUE from desert to subtropical and tropical environments could be explained in terms of differences in evaporation and differences in temperature stress of crops, which are higher in the deserts (Sherwood and Huber, 2010; Tardieu, 2013; Harris and Robinson, 1916). When combined, these two factors have the potential to dramatically depress grain yield and associated WUE (Shah and Paulsen, 2003; Boutraa et al., 2010; Tardieu, 2013). Heat stress and evaporation generally decrease with transition towards tropical and temperate regions. In particular, a lower level of temperature stress under temperate climate may explain the least variation in WUE (CV = 59 %) as compared to the other climates (100, 200 and 90 % for desert, subtropical and tropical, respectively). The low WUE under temperate conditions could be due to the fact that most crop growth cycles are quite long due to low metabolic rates and that not all of the rain is used by crops or evaporated, a significant proportion being infiltrated to the water table (Hendrickson et al., 2004). It was interesting to note field crops produced in the desert climate characterised by generally dry soils and high air temperatures still had significantly higher WUE than crops produced in the temperate climate. Moisture stress in the early growth stages of cereals is known to enhance deep rooting (Çakir, 2004; Chemura et al., 2014) essential for anchorage and reaching out to deeper horizons for soil moisture and nutrients, which could explain the greater WUE in the desert climate. Also, irrigation is provided when the crop requires it and using technologies that decrease water losses. In support, Chibarabada et al. (2015), evaluating the effect of watering regime on groundnut WUE, reported higher WUE under water-stressed than well-watered conditions.

4.2.2. Soil texture

The present study confirms the impact of soil texture on WUE (Mojid et al., 2012). Soil clay content appeared to have a strong negative impact on crop WUE ($r = -0.78$, $p < 0.01$) (Table 5) with high clay soil contents tending to depress crop yields. This trend was, however, invalidated for maize most likely because maize does well in most soil types (Bennetzen and Hake, 2008). High clay contents are most of the time associated with waterlogging conditions and low soil water holding capacity, both detrimental to crop yield (Greenway et al., 2006; Morales-Olmedo et al., 2015). Another interesting result was the trend for WUE to be highest under both silt-loam and loam textures (Fig. 8A, C), which are known to be associated with fertile soils (Russel, 2002).

5. Conclusions

The study investigated the impacts of soil and environmental conditions on water use efficiency (WUE) of main field crops. The first main conclusion using data from 514 sites across the globe was that crops differed significantly in their abilities to efficiently use water for grain yield, with C4 plants such as maize exhibiting significantly higher WUE than C3 types (e.g. wheat). The second conclusion was climate had significant impact on water use efficiency with a tendency for warm and wet climates to yield higher crop WUE than cooler environments. The third conclusion was that soil texture significantly impacted crop WUE with lowest values occurring in clayey soils and highest values under loamy conditions. However, maize performed very well in all soil textures. There is still a certain level of unexplained variability in crop WUE that could potentially be attributed to factors not considered in the present study, which are seldom reported in literature on WUE such as seasonal variations of climate, land management and agronomic practices such as fertilization application, tillage and crop residue mulching. The knowledge generated in the framework of the present study is expected to provide useful insights for policy and

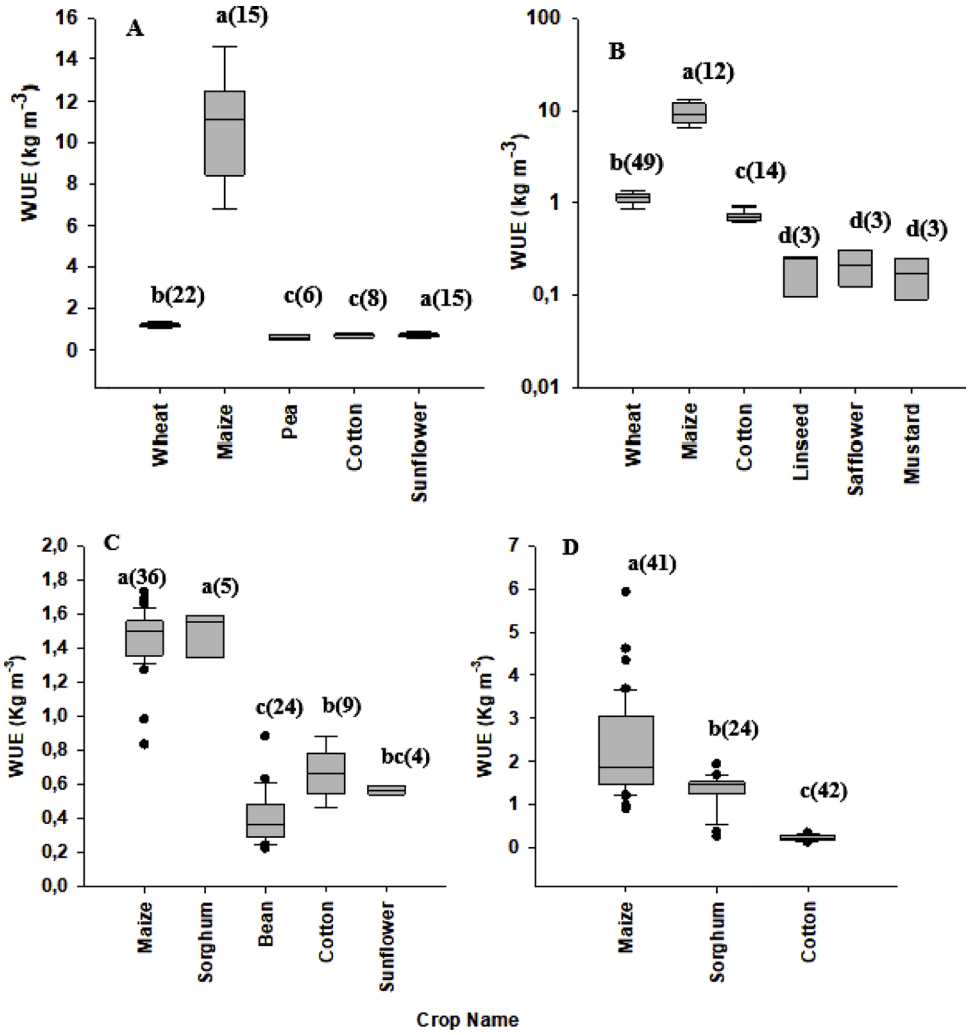


Fig. 8. Box plots comparing WUE (kg m^{-3}) variations for field crops under different soil textures (A) Clay, (B) Loam, (C) Silt-loam and, (D) Clay loam. Box plots accompanied by similar letters were not significantly different at $p < 0.05$. Numbers in brackets are the sample sizes.

decision making in the context of climate uncertainties. The results also suggest that more still needs to be understood about the main controls of field crop water use efficiency. For instance, there seems to exist many interactions between crop water and nutrient availability with for instance Zhang et al. (2017) or Sinclair (2018) to increased WUE following adequate supply, we couldn't investigate in the framework of the present study because of lack of data, that further research needs to consider.

Declaration of Competing Interest

We have no conflict of interest

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